



CAFÉ AROMA

FULL-STACK CAFÉ MANAGEMENT SYSTEM

Team 6 | SE1 Final Project

December 18, 2025

Project Idea

What is Cafe Aroma? | Stakeholders Overview

A role-based system that makes café operations smoother for everyone, from customers placing orders to admins tracking performance.

MOTIVATION FOR THE PROJECT:

One of our team members is passionate about coffee and plans of opening their own café someday, this project let's us build something they could actually use.

STAKEHOLDERS

CUS

Customer

Browse Menu, Order items, Track order status

STA

Staff

Process Orders, Update Status

MAN

Manager

Manage Staff, Menu

ADM

Admin

Manage Manager, Manage Staff, Menu

Our Team

Development Team and Responsibilities



Eric

Project Manager, DevOps & GitHub, Automated Testing



Hedi

DB Setup, Project Setup, Manager Foundation



Anthony

Authentication & Accounts, Manager Foundation



Federico

UI & Page Foundation, Customer Foundation, Staff Foundation



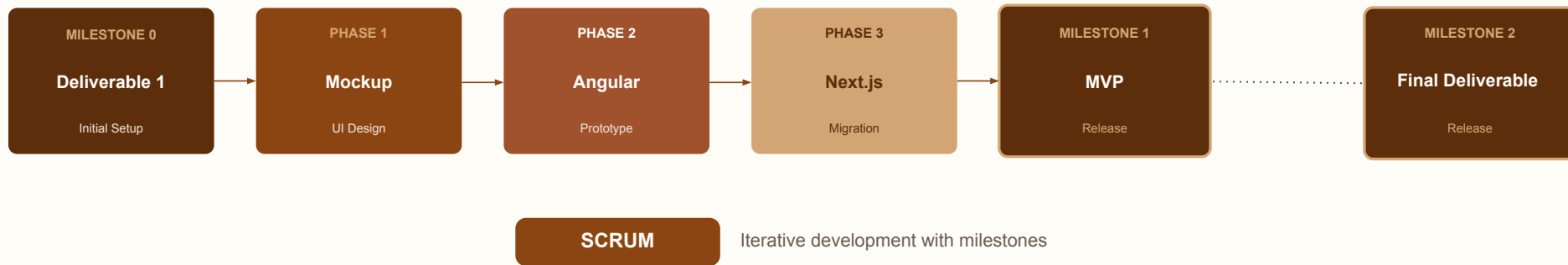
Filip

Customer Foundation

Work	Assignee
 GSA-167 Automated Testing Strategy	 Eric Damian
 GSA-117 Manager Foundation	 Hedi Tebourbi
 GSA-113 Staff Foundation	 Federico Newton
 GSA-79 Customer Foundation	 filip
 GSA-62 UI & Pages Foundation	 Federico Newton
 GSA-61 DevOps & Git Setup	 Eric Damian
 GSA-12 Authentication & Accounts (Supabase Auth)	 Anthony S
 GSA-1 Project Codebase Setup	 Hedi Tebourbi

Development Timeline

Agile/Scrum Approach with Milestones



IMPLEMENTATION ORDER



COORDINATION

- JIRA** Jira board management
- AGILE** Sprint planning
- PR** Code reviews

4. Requirements

MVP Specific Requirements

60 % of ALL REQUIREMENTS

- Shopping Cart & Checkout (FR-CUS-3, FR-CUS-4*)
- Order Queue & Status Management (FR-STA-1, FR-STA-2)
- Menu Management Dashboard (FR-MGR-1)
- Role-Based Access Control (FR-MGR-2)
- Staff order queue with real-time updates
- Order tracking & History (FR-CUS-6)*
- Menu Browsing with Filters (FR-CUS-1)*

NON-FUNCTIONAL REQUIREMENTS



SECURITY

HTTPS, RLS and RBAC implementation



PERFORMANCE

Optimized and low latency using edge deployment and efficient queries



SCALABILITY

Managed Services | Cloud-based (Supabase + Vercel)



DATA INTEGRITY

Ensures consistency through validation and controlled access

Use Cases - Ordering System

MVP Specific Use Cases

CORE USE CASES WE ACHIEVED

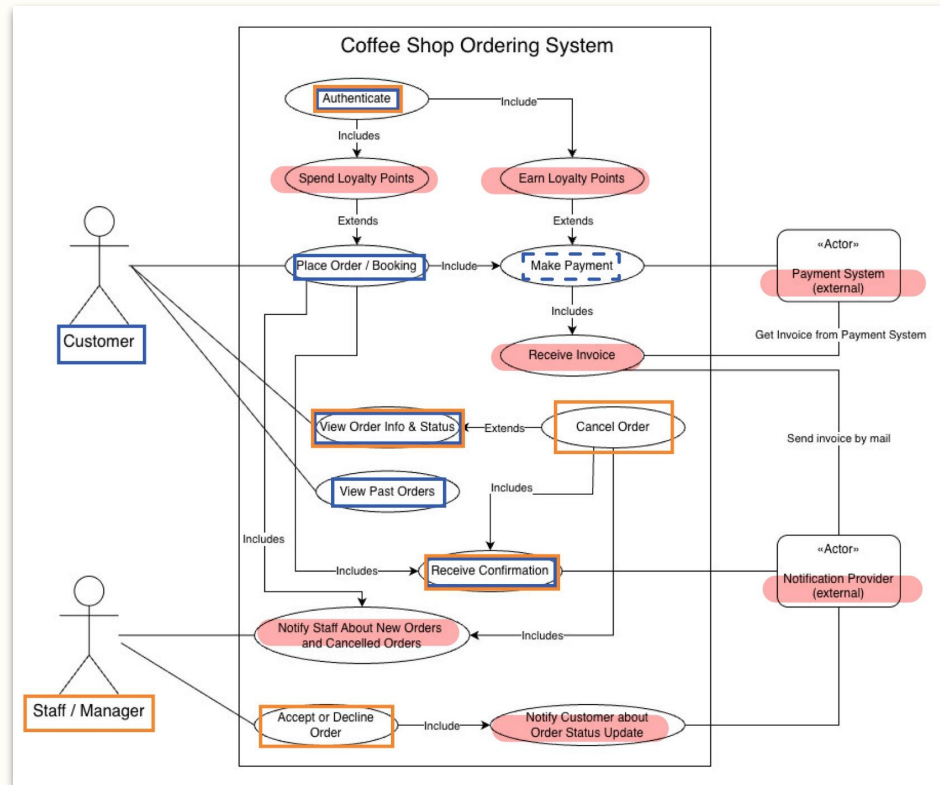
CUSTOMER UC

- Authentication
- Placing an Order Flow
- Order History and Confirmation

STAFF / MANAGER UC

- View Order Information
- Accept an Order
- Order Cancellation
- + Change Order Status

MVP LIMITATIONS



Use Cases - Auth System

MVP Specific Use Cases

CORE UC ALL ACHIEVED (ALMOST)

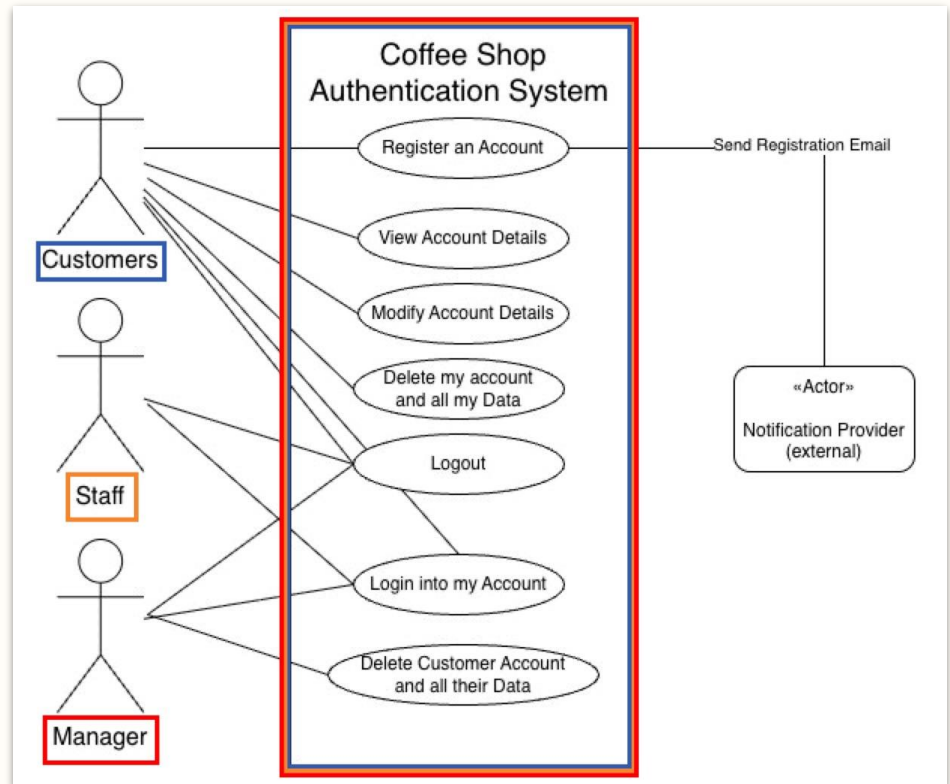
CUSTOMER UC

- Full Authentication Flow
- Register, Log in, Log out, Deletion
- Details Edition

STAFF / MANAGER UC

- Full Authentication Flow
- Special Registering
- Customer deletion not implemented

**NO MAIL
VERIFICATION**



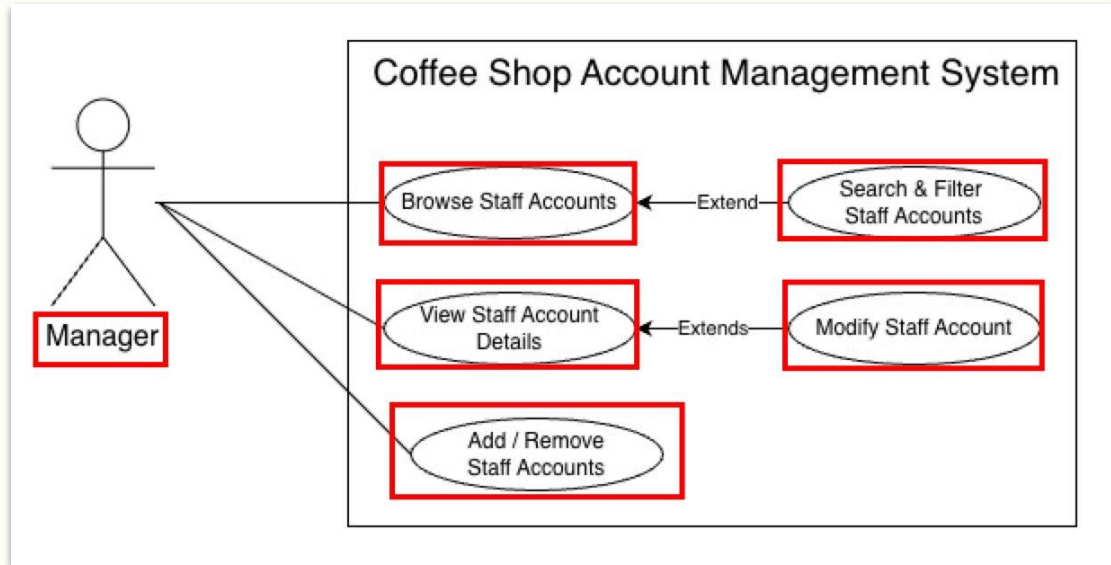
Use Cases - Account Management System

MVP Specific Use Cases

CORE USE CASES WE ACHIEVED

MANAGER

- Browse Staff Accounts (with search and filtering)
- View Details and Modifications
- Add / Remove Staff Accounts
- + Change Order Status



Use Cases - Listing System

MVP Specific Use Cases

CORE USE CASES WE ACHIEVED

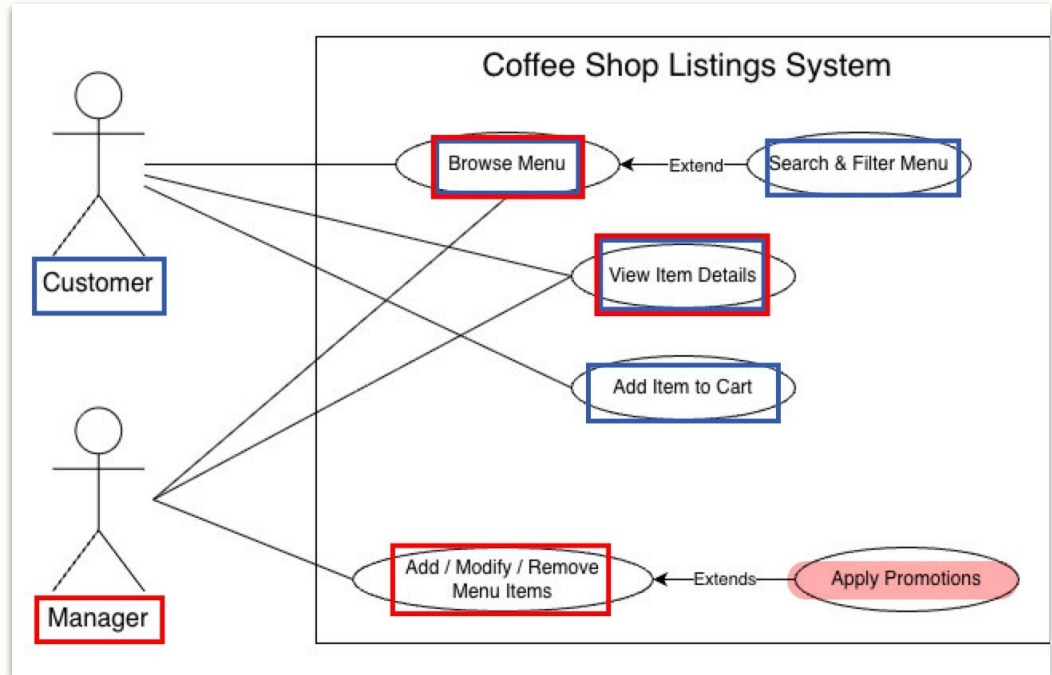
CUSTOMER UC

- Browse Customer Menu
- Search and Filter
- Cart logic

MANAGER UC

- Browse Manager Menu
- Modify Menu
- View Item Details
- Edit Item Details

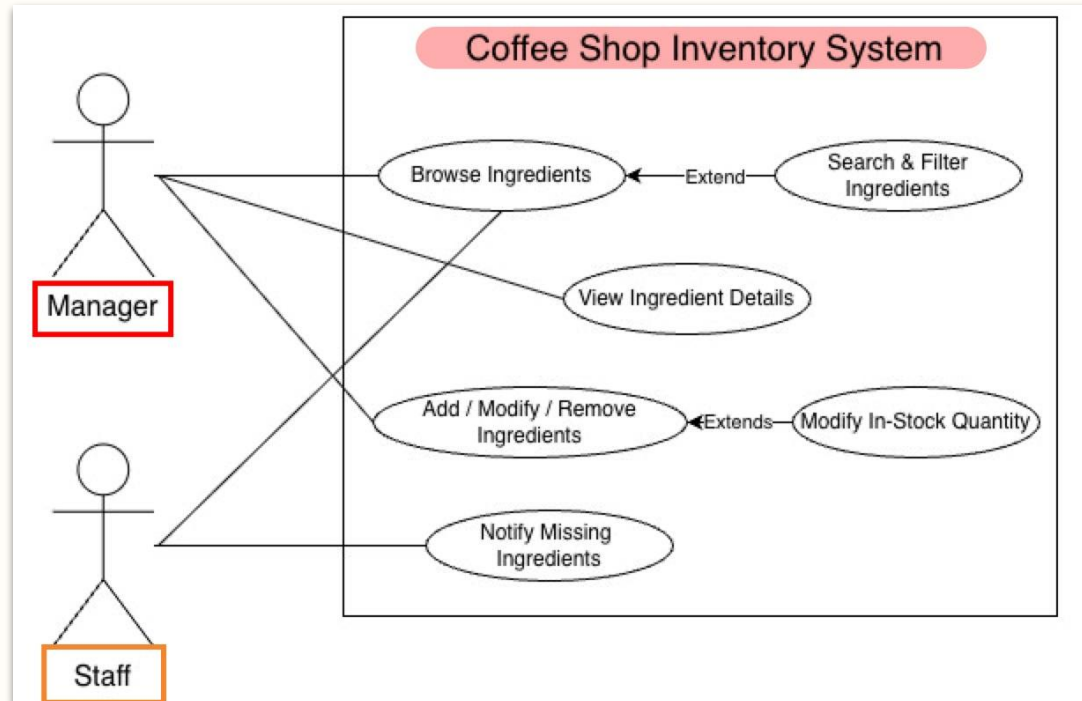
NO PROMOTIONS



Use Cases - Ingredient Inventory System

Not Vital for MVP

INGREDIENTS IMPLEMENTATION IS NOT PART OF THE MVP



Work Organization - Jira

Agile/Scrum Approach Based on Epics

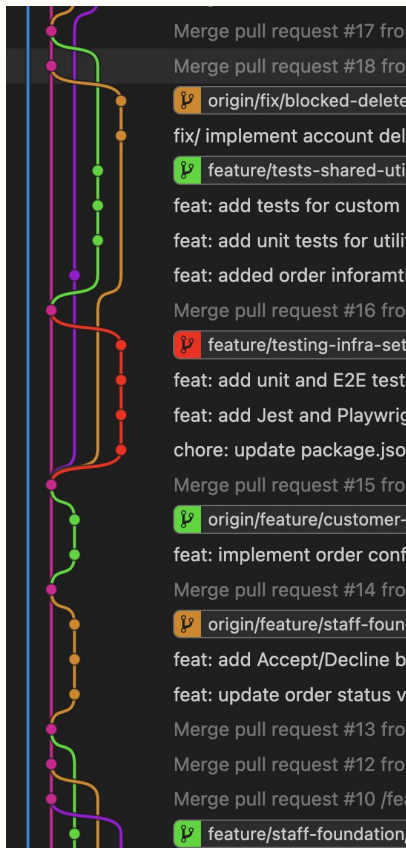
Work	Assignee
🔗 GSA-167 Automated Testing Strategy	 Eric Damian
🔗 GSA-117 Manager Foundation	 Hedi Tebourbi
🔗 GSA-113 Staff Foundation	 Federico Newton
🔗 GSA-79 Customer Foundation	 filip
🔗 GSA-62 UI & Pages Foundation	 Federico Newton
🔗 GSA-61 DevOps & Git Setup	 Eric Damian
🔗 GSA-12 Authentication & Accounts (Supabase Auth)	 Anthony S
🔗 GSA-1 Project Codebase Setup	 Hedi Tebourbi

Report: CSA Sprint 1

Key ↑	Summary	Work type
CSA-2 🔗	Initialize Next.js project	☒ Task
CSA-3 🔗	Configure Tailwind, ESLint, Prettier, Husky	☒ Task
CSA-4 🔗	Create folder structure & UI setup	☒ Task
CSA-5 🔗	Create GitHub repo	☒ Task
CSA-6 🔗	Add PR template + branch protection rules	☒ Task
CSA-7 🔗	Add CI pipeline (build, lint, type-check)	☒ Task

Codebase Management - GitHub

Version Control, PRs, Code Review, and Branch Protection



0 Open ✓ 28 Closed Author ▾

- Hotfix/ci cache clear fix ✓
#28 by devian02 was merged 7 minutes ago • Approved
- Feature/e2e tests ✓
#27 by devian02 was merged 26 minutes ago • Approved 15 of 23 tasks
- feat: consolidate admin and manager dashboards to /manager/dashboard ✗
#26 by RedPill47 was merged 29 minutes ago • Approved
- Feature/tests api routes ✗
#25 by devian02 was merged 1 hour ago • Approved 18 tasks done
- hotfix: allow managers to update orders ✓
#24 by RedPill47 was merged 2 hours ago • Approved
- fix: resolve 401 error when updating customer name ✓
#23 by RedPill47 was merged 2 hours ago • Approved
- Hotfix/copy id button ✓
#22 by Fiizek04 was merged 3 hours ago • Approved 5 tasks done

This branch was successfully deployed
1 active deployment

Review required
At least 1 approving review is required by reviewers with write access.

All checks have passed
5 successful checks

Merging is blocked
New changes require approval from someone other than the last pusher.

Merge pull request ▾ You can also merge this with the command line. [View com](#)

fede-oss commented 4 days ago Member ▾

Reaching the checkout page from the cart works fine, but I cannot manage to create an order as a guest (or account), using cash or card. "Failed to create order. Please try again." Fix and reopen PR

fede-oss closed this 4 days ago

Fiizek04 reopened this 2 days ago

fix(checkout): resolve auth detection for authenticated orders 7712efd

Fiizek04 commented 2 days ago Member Author ▾

@fede-oss, I have pushed a fix for the "Failed to create order" issue you encountered.

The Issue: The server-side Supabase client in `/api/orders/route.ts` was using a shared `getClient()` helper that didn't properly read session cookies in API routes. This caused `auth.getUser()` to return null even for authenticated users, incorrectly treating them as guests.

The Fix: I updated `app/api/orders/route.ts` to create the Supabase client inline with explicit cookie handling (using `get/set/remove` methods), matching the pattern used in the working `/api/auth/user` route. Additionally, I updated `app/checkout/page.tsx` to perform a fresh auth check at submit time to avoid stale React state.

Verification: I performed a full pass of the checkout flow (Guest & Auth) in the preview environment, and order creation is now succeeding.

Ready for a re-review!

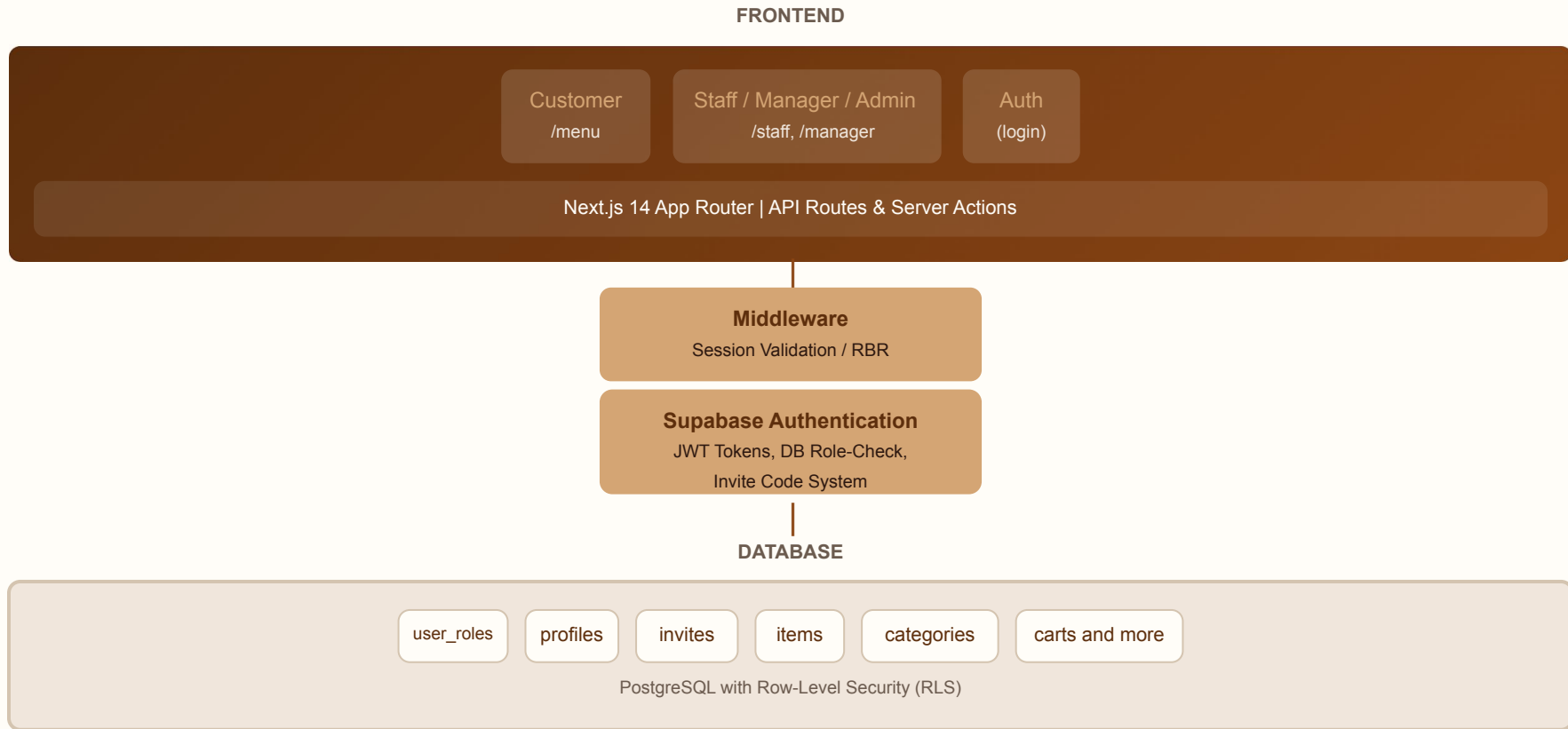
Pull request successfully merged and closed

You're all set — the `feature/customer-foundation/place-order-and-make-payment` branch can be safely deleted.

Delete branch

System Architecture

High-Level Overview



Technology Stack

Modern full-stack architecture

FRONTEND

Next.js 14

App Router, SSR

React 18

Server + Client Components

TypeScript

Type-safe development

Tailwind CSS

Utility-first styling

BACKEND

Next.js API Routes

Server-side endpoints

PostgreSQL with Supabase

Relational database, BaaS platform

Supabase Auth

Email/Password auth

Supabase Realtime

Live order updates (WebSockets)

DEVOPS

Vercel

Frontend deployment

Supabase Cloud

Managed backend

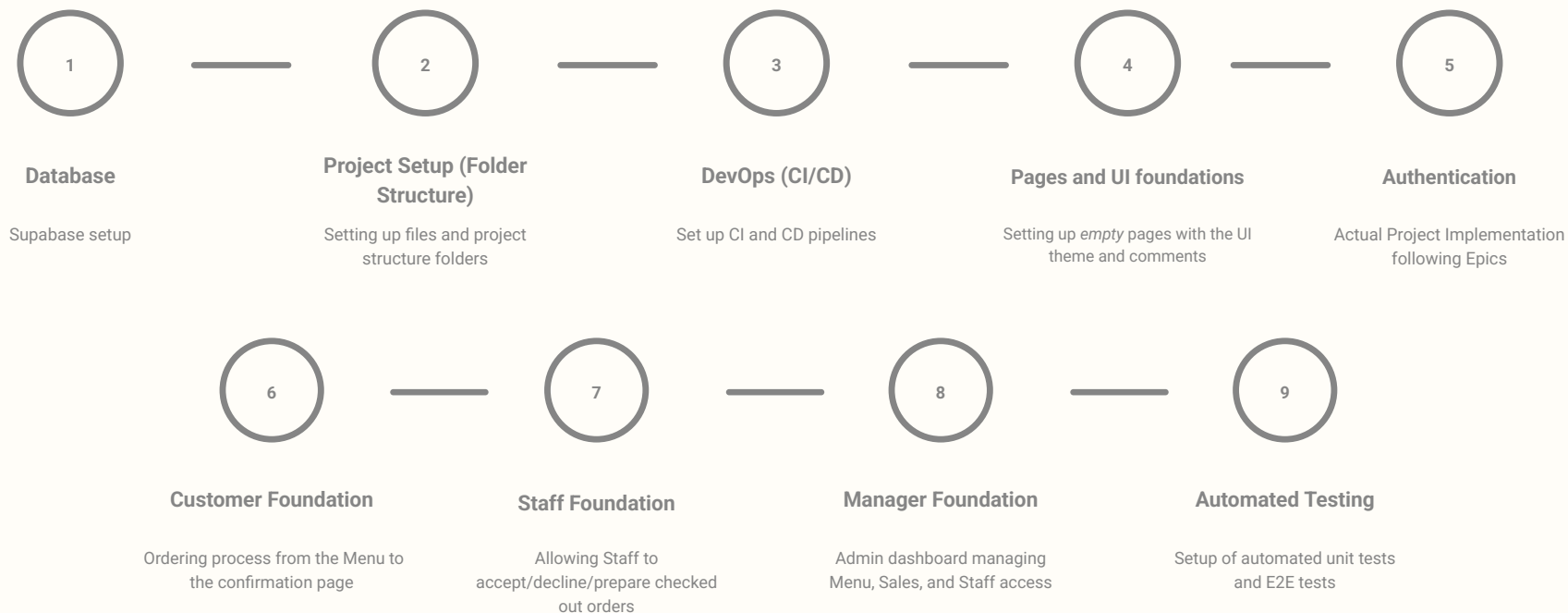
GitHub Actions

CI/CD pipeline

Git

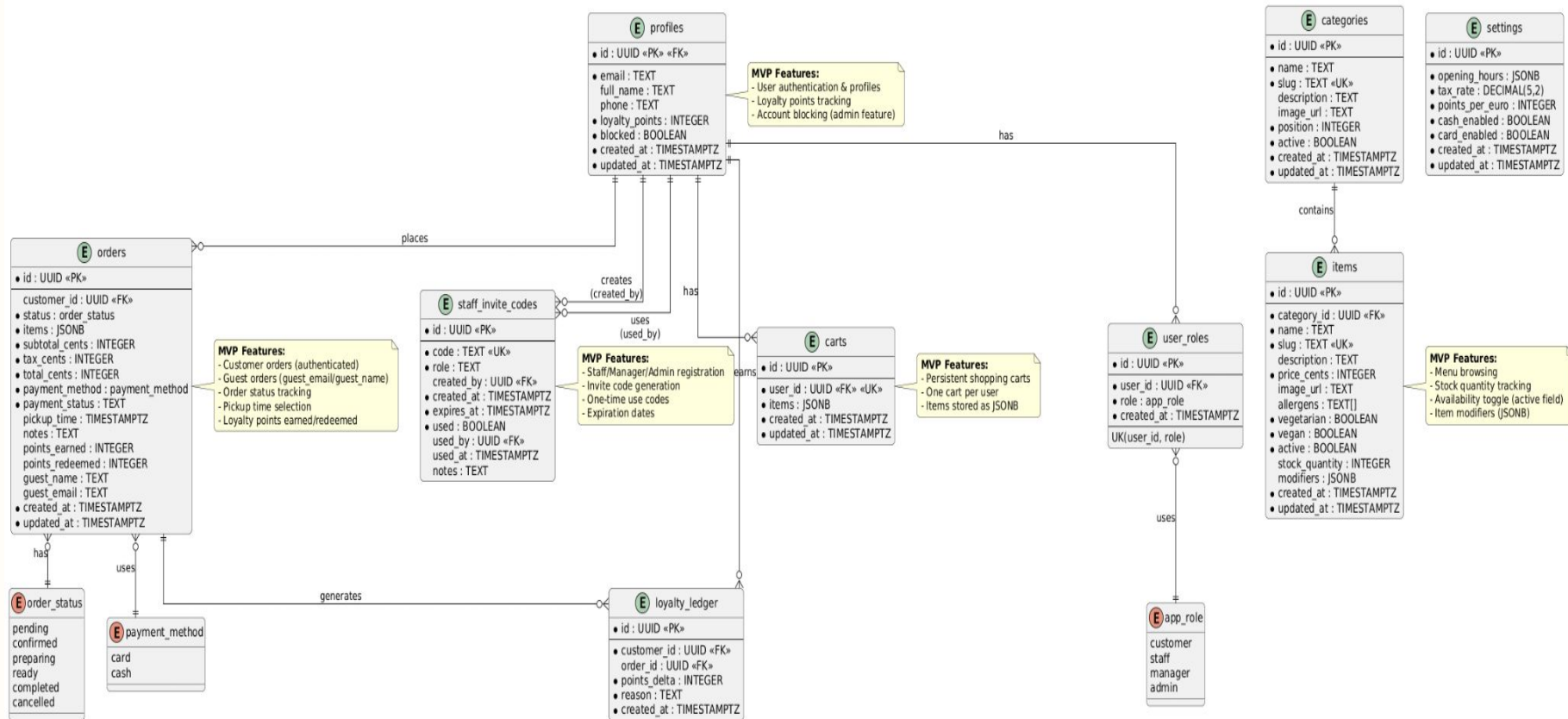
Feature branch workflow

Implementation Timeline



Database Schema

Key tables and relationships



Security Architecture

Authentication and Authorization



Authentication

Supabase Auth handles user login with secure email/password. JWT tokens manage sessions.



Role-Based Access

ADMIN

Full system access

MANAGER

Operations, refunds

STAFF

Order queue, status

CUSTOMER

Own data only

RLS

Row-Level Security

Database-level access control at the PostgreSQL layer.

User Data Isolation

Customers access only own data

Staff Access Control

View orders, update status only

Admin Full Access

Unrestricted table access

OK

API Security: Server-side validation, secure env vars

Epic: Project Setup

- Initialize **Next.js** project
- Create **folder structure** & UI setup
- Create **GitHub repository**
- Set up **Supabase** project
- Configure **environment** variables
- Create database & **RLS** policies

Work		Assignee	
☒	CSA-2 Initialize Next.js project		Hedi Tebourbi
☒	CSA-3 Configure Tailwind, ESLint, Prettier, Husky		Hedi Tebourbi
☒	CSA-4 Create folder structure & UI setup		Hedi Tebourbi
☒	CSA-5 Create GitHub repo		Hedi Tebourbi
☒	CSA-9 Create Supabase project		Hedi Tebourbi
☒	CSA-10 Add environment variables		Hedi Tebourbi
☒	CSA-11 Create initial DB + RLS + migration files		Hedi Tebourbi

Epic: UI pages and foundation

- **Page Structure:** Implemented 15+ pages for customer, manager, and staff with proper routing and semantic HTML
- **Navbar:** with Navigation System responsive dropdowns
- **Modal:** cart modal + menu ready for feature implementation
- **Scalable Foundation:** comments with placeholder content with Next.js conventions
- **Design:** “Coffee-style” theme for all pages

Work	Assignee
☒ GSA-63 Implement base app layout (header, main, footer)	 Federico Newton
☒ GSA-64 Implement global navigation structure & route links	 Federico Newton
☒ GSA-65 Create /menu page	 Federico Newton
☒ GSA-66 Create /auth/login page	 Federico Newton
☒ GSA-67 Create /auth/register page	 Federico Newton
☒ GSA-68 Create /account page	 Federico Newton
☒ GSA-69 Create Cart Modal component	 Federico Newton
☒ GSA-70 Create /checkout page	 Federico Newton
☒ GSA-71 Create /customer/orders page	 Federico Newton
☒ GSA-72 Create /customer/account page	 Federico Newton
☒ GSA-73 Create /manager/dashboard page	 Federico Newton
☒ GSA-74 Create /manager/staff-management page	 Federico Newton
☒ GSA-75 Create /manager/menu page	 Federico Newton
☒ GSA-76 Create Manager Menu Item editor modal	 Federico Newton
☒ GSA-77 Create /staff/menu page	 Federico Newton
☒ GSA-78 Create /staff/orders page	 Federico Newton

Epic: Authentication & Accounts (Supabase Auth)

- **Customer registration:** Form → API → signUp() → TriggerDB
- **Staff/Admin signup:** staff_invite_codes; role, expiration, single-use. Trigger checks valid → assigns role → Marks as used
- **Logout button** clears session completely, kills server session
- **View account details** email, name, role, status
- **Update profile** only name for now → cannot modify own role
- **Delete account** Service_Role_Key → Bypasses RLS
- **Role-based routing:** Queries Middleware → Checks for Blocked, Each Role gets redirected to the right side of the frontend
- **Persistent session** with useUser() hook + listener, JWT
- **Blocked users** redirected to dedicated page

✓ ⚡ CSA-12	Authentication & Accounts (Supabase Aut...	AS Anthony S
> 📖 CSA-57	Blocked user handling	AS Anthony S
> 📖 CSA-53	Persistent session	AS Anthony S
> 📖 CSA-48	Role-based access / correct dashboard	AS Anthony S
> 📖 CSA-43	Delete account & data	AS Anthony S
> 📖 CSA-38	Modify account details	AS Anthony S
> 📖 CSA-33	View account details	AS Anthony S
> 📖 CSA-29	Log out	AS Anthony S
> 📖 CSA-24	Log in	AS Anthony S
> 📖 CSA-19	Staff/Admin signup	AS Anthony S
> 📖 CSA-13	Register an account	AS Anthony S

Epic: DevOps & Git Setup

- Created **PR template** and configured **branch protection rules** for code review enforcement
- Implemented **CI pipeline** with automated build, automated testing, linting, and type-checking workflows
- Configured **Vercel deployment** for automated hosting and preview environments

✓ GSA-61 DevOps & Git Setup	Eric Damian
✓ GSA-8 Setup Vercel deployment	Eric Damian
✓ GSA-7 Add CI pipeline (build, lint, type-check)	Eric Damian
✓ GSA-6 Add PR template + branch protection rules	Eric Damian

Branching Strategy	
Never commit directly to <code>main</code> . All development happens in <code>dev</code> .	
Branch Prefix	Purpose
<code>feature/</code>	new features
<code>refactor/</code>	restructures & optimizations
<code>bugfix/</code>	fixes for known issues
<code>hotfix/</code>	urgent production fixes
<code>chore/</code>	maintenance work

ci	
>	✓ Set up job
>	✓ Run actions/checkout@v4
>	✓ Setup Node
>	✓ Install dependencies
>	✓ Lint
>	✓ Type check
>	✓ Run Unit & Integration Tests
>	✓ Upload Test Coverage
>	✓ Build
>	✓ Post Setup Node
>	✓ Post Run actions/checkout@v4
>	✓ Complete job



This branch was successfully deployed

1 active deployment



Review required

At least 1 approving review is required by reviewers with write access.



All checks have passed

5 successful checks



Merging is blocked

New changes require approval from someone other than the last pusher.

Merge pull request

You can also merge this with the command line. [View command](#)

Deployments		
Automatically created for pushes to Software-engineering-1-U		
6nUxuKedS Production	Ready 15s	main >_ 129b7ba
30yP3BkN9 Production	Ready 1m 18s	main ~> 129b7ba
Dtez2Jkc3 Preview	Ready 15s	dev >_ c884708
7oEYS9HPA Preview	Ready 45s	dev ~> c884708
2M9UVRTYG Preview	Ready 18s	dev >_ a6af524
DXQcjVpOd Preview	Ready 55s	dev ~> a6af524

Epic: Customer Foundation

- **Browse menu** items available at the coffee shop
- **Add** selected items **to shopping cart** for purchase
- **Place order** and make payment as guest or user
- Receive **order confirmation** immediately after successful checkout
- **View order details** and current status anytime
- **Receive** and **download invoice** for completed orders
- **View** complete **order history** and past purchases

☐	Work	Assignee
☐	▼ ⚡ GSA-79 Customer Foundation	F filip
☐	> 📌 GSA-164 View orders history ↗ +	F filip
☐	> 📌 GSA-85 Receive invoice	F filip
☐	> 📌 GSA-84 View order info & status (also as guest)	F filip
☐	> 📌 GSA-83 Receive order confirmation	F filip
☐	> 📌 GSA-82 Place an order and make payment (also as guest)	F filip
☐	> 📌 GSA-82 Place an order and make payment (also as guest)	F filip
☐	> 📌 GSA-81 Add items to cart	FN Federico Newton
☐	> 📌 GSA-80 Browse menu ↗ +	FN Federico Newton

Epic: Staff Foundation

- **See Menu + Availability:** Stock levels with availability badges, category filters
- **Staff Order Queue:** Real-time queue, priority sorting, status filters, high-priority badges
- **Accept, Decline, Complete Orders:**
Accept/Decline/Complete buttons, confirmation modals, status update API

▼ ⚡ GSA-113 Staff Foundation	↗ +	 Federico Newton
> 📌 GSA-116 Accept, decline or complete orders		 Federico Newton
> 📌 GSA-115 Order queue with priorities		 Federico Newton
> 📌 GSA-114 See current menu & availability		 Federico Newton

Epic: Manager Foundation

- **Edit menu items**; add new products and set prices
- **View** analytics, metrics, recent orders and alerts
- **See all orders**; filter, sort by priority
- **List** all staff **accounts**; search and filter
- View **staff details**; edit name, email, role
- **Create staff** invite **codes**; block/unblock accounts

▼ ⚡ GSA-117 Manager Foundation	HT Hedi Tebourbi
> 📌 GSA-134 Add or remove staff accounts	AS Anthony S
> 📌 GSA-133 View & edit staff account details	AS Anthony S
> 📌 GSA-132 Browse staff accounts	AS Anthony S
> 📌 GSA-131 Orders page with priorities	HT Hedi Tebourbi
> 📌 GSA-130 Manager metrics dashboard	HT Hedi Tebourbi
> 📌 GSA-129 Browse & manage menu items	HT Hedi Tebourbi

Epic: Automated Testing Strategy

- Set up **testing infrastructure** with **Jest**, **Playwright**, and **coverage reporting**
- Established **unit tests** for shared utilities and custom React hooks
- Developed **component tests** for reusable UI components using **React Testing Library**
- Created comprehensive API route and **server logic tests with Jest**
- Implemented **end-to-end tests** for critical user flows using Playwright
- **Integrated automated testing into CI/CD pipeline** for continuous quality checks

▼ ⚡ GSA-167	Automated Testing Strategy	ED Eric Damian
> 📌 GSA-190	CI/CD Integration for Testing	ED Eric Damian
> 📌 GSA-186	Critical User Flow End-to-End Tests	ED Eric Damian
> 📌 GSA-183	API Route & Server Logic Testing	ED Eric Damian
> 📌 GSA-179	Component Testing for Reusable UI Comp...	ED Eric Damian
> 📌 GSA-175	Unit Testing for Shared Utilities & Hooks	ED Eric Damian
> 📌 GSA-168	Testing Infrastructure Setup	ED Eric Damian

ci 🔍 ⌂ ⚙️

▼ ✓ Run Unit & Integration Tests 15s

1473
1474 Test Suites: 29 passed, 29 total
1475 Tests: 466 passed, 466 total
1476 Snapshots: 0 total
1477 Time: 11.95 s
1478 Ran all test suites.

> ✓ Upload Test Coverage 4s

Outlook & Future

Next steps and future enhancements

CURRENT STATUS

MVP complete with all core features implemented. Application deployed and functional.

TIMELINE

● Today (Dec 18)

Final presentation & demo

● Dec - Jan

Addition of the Final Use Cases, Bug fixes, polish

● Feb 4, 2026

Final submission deadline

FUTURE ENHANCEMENTS

Mobile responsive design

Enhance the mobile experience

Mail notifications

Order ready alerts

Potential Payment Support

Stripe, PayPal support

Improved Inventory Tracking

Stock management for ingredients, alerts

Analytics Dashboard Improvement

Sales reports, insights

Loyalty System

Tiers, rewards redemption



Live MVP Demonstration

10 Minutes



Customer

3-4 min

-



Staff

2-3 min

-



Manager + Admin

2-3 min

Check it yourself: <https://se-team-6-project-coffee-app.vercel.app/>

Thank You!

